

Cost model: LangSmith vs Langfuse

A worked cost comparison for a real workload, with every line itemised and every assumption stated.

Headline finding: at this volume you are not buying traces, you are buying seats. Trace ingestion costs between \$0 and \$1.50 a month. Everything else on the bill is per-user pricing.

What you are actually paying for

The two vendors meter completely different things, which is why a naive price comparison misleads.

LangSmith bills on five dimensions

Dimension	Rate	Notes
Seats	\$39 / seat / month (Plus)	Developer is free but capped at 1 seat
Base traces	\$0.50 per 1,000	14-day retention. 5k included on Developer, 10k on Plus
Extended traces	\$5.00 per 1,000	400-day retention, 10x the base rate
Engine	\$1.50 per LCU	LangChain Compute Unit. Consumption not published
Insights	~\$1 to \$2 per 1,000 threads	OpenAI models. \$3 to \$4 with Anthropic

Deployments, Sandboxes and Fleet are also metered in LCU and LSU (\$1.00 per LangChain Storage Unit), but they are not part of an observability decision.

Langfuse bills on one dimension

Dimension	Rate
Units (roughly one span)	Graduated: \$8 per 100k up to 1M, then \$7 , then \$6.50 , then \$6
Seats	Unlimited on every paid plan. Hobby is capped at 2 users

Plan floors: Hobby \$0 (50k units, 30-day retention), Core \$29 (100k units, 90 days), Pro \$199 (100k units, 3-year retention), Enterprise \$2,499.

The unit conversion that matters

Langfuse units are not traces. This lab measured it directly: **20 traces produced 198 observations**, so this agent averages **9.9 units per trace**. Every Langfuse figure below uses that multiplier. An agent with more tool calls will be worse; a single-shot completion will be better.

The scenario

300 to 400 traces per day, which is 9,000 to 12,000 traces per month, from a team of engineers who all need access to the tool.

	300/day	400/day
Traces per month	9,000	12,000
Langfuse units per month	89,100	118,800
Storage growth, self-hosted	65 MB/month	87 MB/month

Note the storage line. At this volume, disk is irrelevant. The 100 GB Langfuse recommends would last decades.

Cost build-up at 400 traces/day

LangSmith

Trace cost is effectively zero. 12,000 traces against a 10,000 allowance leaves 2,000 billable at \$0.50 per thousand, so **\$1.00 a month**.

Team size	Seats	Traces	Total	Seats as % of bill
1 engineer	\$39	\$1.00	\$40	98%
3 engineers	\$117	\$1.00	\$118	99%
5 engineers	\$195	\$1.00	\$196	99.5%
10 engineers	\$390	\$1.00	\$391	99.7%

Add the optional intelligence features:

Feature	Cost at this volume
Insights	\$12 to \$24/month if you analyse all 12,000 threads with OpenAI models. Less if you sample
Engine	Cannot be forecast. See the warning below

[!WARNING] **Engine cost is genuinely unpredictable.** It is metered at \$1.50 per LCU, and LangChain publishes no LCU-per-trace consumption figure. Their documentation says usage "depends on the number of traces Engine analyses, the depth of analysis required, and the amount of work it performs". If you plan to use Engine, get a consumption estimate from LangChain before committing. Do not model it from list price alone.

The Developer plan is worth knowing about. It is free, includes 5,000 traces, and would cost **\$3.50/month** at this volume. But it is capped at **one seat** and excludes Engine and Insights entirely. It is a single-developer plan, not a team plan.

Langfuse Cloud

Line	Cost
Core plan base	\$29.00
Units: 118,800 against 100,000 included, 18,800 over at \$8/100k	\$1.50
Seats	\$0, unlimited
Total	\$30.50/month

Flat regardless of team size. Ten engineers cost the same as one.

Langfuse self-hosted

Treated as an independent infrastructure cost, since that is what it is.

Line	Monthly	Basis
Compute: 1 VM, 4 vCPU / 16 GiB	~\$120	Langfuse's recommended sizing, at published on-demand rates
Storage: 100 GB block	~\$10	Lasts decades at this volume
Backups and snapshots	~\$5	
Langfuse licence	\$0	MIT. All core features
Infrastructure subtotal	~\$135	
Operations	see below	The line that actually decides this

The measured resource profile backs the sizing. From [05-scale.md](#): the stack idles at **2.16 GiB** and peaks at **3.2 GiB** under a 395 traces/second load. At 400 traces per **day** you are using a fraction of one percent of that capacity. A 4 vCPU / 16 GiB box is generous, and the recommendation exists for production ingest, not for this.

Reserved instances or a committed-use discount typically cut the compute line by 30 to 60 percent. A small always-on VM from a budget provider can be under \$40.

Side by side, 400 traces/day, 5 engineers

Option	Monthly	Seats	Data location
LangSmith Developer	\$3.50	1 only	Their cloud
Langfuse Cloud Core	\$30.50	Unlimited	Their cloud
Langfuse self-hosted, infra only	~\$135	Unlimited	Your network
LangSmith Plus	\$196	5	Their cloud
LangSmith Plus + Insights	~\$214	5	Their cloud
Langfuse self-hosted, infra + ops	~\$535 to \$935	Unlimited	Your network

Two things jump out.

Langfuse Cloud is roughly one sixth the price of LangSmith Plus at this team size, and the gap widens with every engineer you add.

Self-hosting is not the cheap option at this volume. Once you count the person who runs it, it is the most expensive line on the page. Self-hosting is a data-residency decision, not a cost-saving one, until you are at far higher volume.

The operations line

This is the number that gets left off vendor comparisons, and it dominates everything else at this scale.

A single-node Docker Compose deployment realistically needs **4 to 8 hours a month** of someone's attention: version upgrades, ClickHouse disk management, backup verification, certificate renewal, and responding when the worker stops draining.

Loaded engineering rate	4 hrs/month	8 hrs/month
\$50/hour	\$200	\$400
\$100/hour	\$400	\$800

At \$100/hour and 8 hours, **operations costs six times the infrastructure**. Substitute your own rate, but do not leave the line at zero.

Langfuse's own documentation is explicit that Docker Compose "lacks high-availability, scaling capabilities, and backup functionality", and recommends Kubernetes for production. Kubernetes raises both the infrastructure and the operations line.

What changes the answer

Team size

At 400 traces/day, holding volume constant:

Engineers	LangSmith Plus	Langfuse Cloud	Ratio
1	\$40	\$30.50	1.3x
3	\$118	\$30.50	3.9x
5	\$196	\$30.50	6.4x
10	\$391	\$30.50	12.8x
20	\$781	\$30.50	25.6x

Team size is the single biggest cost driver in this comparison. It is worth deciding deliberately who genuinely needs a LangSmith seat rather than buying one per engineer.

Volume

Holding the team at 5 engineers:

Traces/month	LangSmith Plus	Langfuse Cloud	Self-hosted infra	Self-host storage growth
9,000	\$195	\$29	~\$135	0.1 GB/mo
12,000	\$196	\$31	~\$135	0.1 GB/mo
100,000	\$240	\$100	~\$135	0.7 GB/mo
1,000,000	\$690	\$724	~\$135	7.1 GB/mo
10,000,000	\$5,190	\$6,271	~\$350	71 GB/mo

The crossovers:

- **Langfuse Cloud beats LangSmith** until about 1 million traces a month, where the per-unit multiplier catches up with LangSmith's per-trace rate.
- **Self-hosting beats both** from roughly 200,000 traces a month on infrastructure alone, and decisively past 1 million.
- **At 10 million a month, self-hosting is 15 to 18 times cheaper**, and that is where the operations cost genuinely pays for itself.

Storage figures come from the measured **7.27 MB per 1,000 traces** in [05-scale.md](#). At 10M traces a month you add roughly 850 GB a year, so disk becomes a real line item rather than a rounding error.

Retention

Easy to miss and expensive. LangSmith base traces are kept **14 days**. Extended retention is **400 days at 10x the price**, and traces with feedback attached are upgraded automatically, which is exactly what your evaluation runs produce.

At 12,000 traces a month, if a quarter get upgraded that is 3,000 traces at \$5.00 per thousand, so **\$15/month** against \$1.00 for the base tier. Small here, but it scales linearly and it surprises people.

Self-hosted Langfuse keeps everything until you delete it, at 7.27 MB per 1,000 traces.

Recommendation

For **300 to 400 traces a day with a team of engineers**:

1. Langfuse Cloud Core at \$29 to \$31/month is the best value by a wide margin. Unlimited seats, 90-day retention, and no infrastructure to run. It costs less than a single LangSmith seat.

2. Add a small number of LangSmith Plus seats only if the team needs Studio for LangGraph debugging or wants Engine and Insights. Two seats is \$78/month. Buying a seat per engineer at \$39 is where the budget disappears.

3. Self-host Langfuse only if data residency requires it. At this volume it costs more than Langfuse Cloud once operations are counted. It becomes the obvious answer past roughly 1 million traces a month, or immediately if trace data cannot leave your network.

4. Revisit at 1 million traces a month. That is where every crossover in this document sits.

Assumptions and provenance

Being explicit about which numbers are hard and which are estimates.

Measured in this lab: - 9.9 Langfuse units per trace (20 traces produced 198 observations) - 7.27 MB of storage per 1,000 traces - 2.16 GiB idle, 3.2 GiB peak memory for the six-container stack - 395 traces/second end-to-end ingest ceiling on 14 cores

Published by the vendors (August 2026, verify before committing): - LangSmith: \$39/seat, \$0.50 per 1k base traces, \$5.00 per 1k extended, \$1.50/LCU, \$1.00/LSU, Insights at \$1 to \$2 per 1k threads - Langfuse: \$0 / \$29 / \$199 / \$2,499 plan floors, graduated \$8 / \$7 / \$6.50 / \$6 per 100k units, unlimited seats on paid plans, MIT licence for self-hosted

Estimated, and the weakest numbers here: - ~\$120/month for a 4 vCPU / 16 GiB VM, from published on-demand list pricing. Not measured. Reserved pricing cuts this substantially - 4 to 8 hours/month of operations effort for a single-node deployment - 30 days per month for the daily-to-monthly conversion

Not forecastable: - LangSmith Engine consumption. Metered per LCU with no published rate per trace. Get a quote from LangChain

Sources

- [LangSmith pricing](#)
 - [LangSmith Insights documentation](#)
 - [LangSmith Engine](#)
 - [Langfuse pricing](#)
 - [Langfuse self-hosting pricing](#)
 - [Langfuse Docker Compose sizing](#)
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